A Survey on RAG-Based Multimodal Embedding Techniques

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***Abstract*—In real-world applications, knowledge often exists in multimodal forms , which traditional text-only Retrieval-Augmented Generation (RAG) techniques struggle to handle effectively. To overcome this limitation, recent research has focused on integrating multimodal embedding techniques into RAG systems, enabling models to process diverse data types and improve generation accuracy. This survey provides a comprehensive review of RAG techniques augmented with multimodal embeddings, covering the underlying architecture, key developments, evaluation methodologies, and practical applications. We categorize existing works from multiple perspectives, including their embedding strategies and cross-modal integration mechanisms, and highlight key findings and performance improvements across tasks. In particular, we find that multimodal RAG systems achieve significant gains in tasks like medical image analysis and cross-modal content generation by grounding outputs in relevant retrieved information. Finally, we discuss open challenges and outline future directions for this emerging field.**

***Index Terms*—Survey, Retrieval-Augmented Generation , Multimodal Embedding, Multimodal Evaluation Methods**

I. Introduction

In recent years, Multimodal Large Language Models (MLLMs) have shown impressive performance in diverse domains, prompting exploration of their integration into retrieval-augmented architectures. [1], [2]. In practical scenarios, knowledge often exists in diverse modalities, including text, images, and videos. This diversity necessitates models capable of processing and understanding information from multiple modalities simultaneously. However, traditional text-based Retrieval-Augmented Generation (RAG) techniques [3]–[5] are inherently limited to handling textual data. They struggle with non-text modalities such as images or videos. This fundamental shortcoming constrains their applicability in real-world multimodal tasks. For example, conventional RAG models often fail to effectively leverage visual content in tasks that require joint image-text understanding. As a result, their outputs often lack visual grounding and contextual coherence [6]–[8].

To overcome these limitations, Multimodal Embedding RAG has garnered increasing attention in recent studies [9]. Unlike traditional RAG systems that solely rely on textual retrieval and generation, multimodal RAG integrates data from text, images, videos, and other sources. By employing embedding techniques **embedding techniques** [10], it projects heterogeneous data into a shared vector space. This unified representation enables the model to jointly process and comprehend inputs from multiple modalities. This approach significantly enhances the performance of RAG in tasks such as image captioning, visual question answering, and cross-modal retrieval.

This paper presents a comprehensive review of the development of retrieval-augmented generation techniques with multimodal embedding. The discussion encompasses theoretical foundations, recent advances, and practical applications. We begin by outlining the fundamental concepts and operational mechanisms of RAG. We then highlight its core components—namely the retrieval and generation modules—and explain their collaborative role in improving generation tasks. Subsequently, we delve into recent progress in multimodal embedding RAG from several perspectives:

- We review the evolution of text embedding techniques, including classical models such as Word2Vec and GloVe, as well as context-aware methods like BERT. These approaches have significantly improved the semantic representation and expressiveness of textual data, especially in generative tasks.
- We analyze the Transformer architecture, detailing its self-attention and multi-head attention mechanisms. We explain how these mechanisms contribute to capturing long-range dependencies and global context in multimodal data.
- We explore the role of pretrained language models such as BERT and GPT in multimodal RAG systems. We discuss their strengths and limitations, with particular reference to OpenAI's contributions and these models' performance in generative tasks.
- We describe the integration of retrieval and generation modules through unified embedding spaces, where multimodal data are mapped such that semantically similar content lies in close proximity. This enables more effective cross-modal understanding and generation.
- We discuss task-specific fine-tuning techniques, where pretrained models are adapted to domain-specific tasks to enhance performance in specialized applications.
- We emphasize the importance of the synergy between retrieval and generation modules in multimodal RAG

frameworks. The retrieval component gathers relevant information across text, image, and video modalities, while the generation module utilizes this contextual input to produce accurate and diverse outputs.

Additionally, Section II provides an overview of the core components of RAG. Section IV introduces evaluation metrics for RAG systems, including measures of relevance, accuracy, and faithfulness. We also present emerging evaluation frameworks tailored for multimodal RAG, such as ARES, CRUD, and RAGAS. Finally, Section V examines the practical deployment of multimodal RAG in domains such as medical imaging and artistic content creation. These applications illustrate the potential of multimodal RAG to address real-world multimodal challenges.

To systematically investigate the integration of multimodal embeddings into RAG systems, we conducted a structured literature review across major AI publication databases, covering works from 2020 to 2025. Our goal was to synthesize current architectures, categorize implementation strategies, and identify open research gaps in this emerging field.

II. Overview of Retrieval-Augmented Generation

Retrieval-Augmented Generation models consist of two core components: the **retriever** and the **generator** [11]. The retriever module is responsible for identifying and retrieving relevant documents or passages from a large-scale knowledge base in response to an input query. This retrieval process is typically implemented using traditional or modern information retrieval algorithms, such as BM25, dense retrievers, or other retrieval techniques [12]. The primary objective of the retriever is to ensure that the retrieved content is highly relevant and provides sufficient contextual support for the subsequent generation phase.

The generator module is generally built upon large language models (LLMs), such as those based on the GPT architecture. Its role is to synthesize a coherent and informative response by integrating the retrieved context with the original input query. During this process, the generator must not only comprehend the user query but also effectively leverage the retrieved context to produce outputs that are both fluent and informative. The performance of the generator is largely determined by its ability to utilize the retrieved knowledge and its depth of understanding of the query.

III. Advancements in Multimodal Embedding

A. Text Embedding

Text embedding aims to project natural language text into a high-dimensional vector space, where semantically similar words are located closer to one another [13]. Initially, methods such as Word2Vec and GloVe were widely used to capture statistical relationships between words. However, these approaches relied on static word embeddings and failed to account for contextual variations in word meanings [14].

With the introduction of pretrained language models such as BERT and GPT, text embedding has advanced significantly. These models leverage bidirectional Transformer architectures to capture contextual semantics from both directions of the input, enabling more precise text representations [15]. Such context-aware embeddings have substantially improved the performance of both text generation and comprehension tasks.

B. Transformer Architecture

The Transformer architecture marked a paradigm shift in natural language processing (NLP), offering significant advantages in modeling long-range dependencies through its self-attention mechanism [16]. Self-attention dynamically assigns weights to different parts of the input sequence during encoding, capturing global information effectively. Compared to traditional RNNs and LSTMs, Transformers exhibit superior efficiency and accuracy in complex linguistic tasks [17].

By mapping heterogeneous data types—such as text, images, and videos—into a shared representation space, Transformers enable effective multimodal data fusion. For example, the ViLT (Vision-and-Language Transformer) model adopts a unified Transformer architecture to jointly embed visual and textual inputs, thereby improving performance on multimodal tasks [18].

C. Pretrained Language Models

RAG systems combine retrieval and generation mechanisms to enhance contextual relevance and output accuracy. A key innovation in RAG lies in its dual-stage pipeline: the retriever identifies and extracts relevant content from a large knowledge base, potentially containing data in multiple modalities such as text, images, and video. The generator then leverages this contextual information to produce coherent and informative responses [19]. The system's effectiveness hinges on the synergy between the retrieval and generation components.

Despite notable progress, multimodal RAG still faces several challenges. One major difficulty is the integration of heterogeneous data sources, as different modalities often have fundamentally distinct representations. This discrepancy can hinder retrieval efficiency and degrade output quality. Moreover, the retrieval component must handle vast, often unstructured datasets. While algorithms like BM25 and dense retrievers perform well for text-based retrieval, aligning them with non-textual modalities remains complex.

The generation component must also maintain high output quality. Although advanced models such as GPT-4, PaLM-2, and LLaMA demonstrate strong text generation capabilities, ensuring accurate integration of multimodal context—without introducing inconsistencies or omitting key information—remains a significant challenge. Additionally, scalability and robustness are critical concerns, especially when extending these models to cross-domain, large-scale, and diverse datasets. Addressing these limitations is essential to fully realize the potential of multimodal RAG in real-world applications.

D. Retrieval Module and Generation Model

In RAG frameworks, the integration of the **retrieval module** and the **generation model** is fundamental to enhancing overall generative capabilities [20]. The retrieval module plays a

crucial role by extracting relevant content from large-scale external knowledge bases to provide necessary context for generation. By sourcing information from diverse datasets, it ensures that the generation model has access to high-quality, contextually appropriate content. This enables the model to produce more accurate, informative, and trustworthy outputs, particularly when external knowledge beyond the training data is needed.

Retrieval can be based on various techniques, including **Content-Based Retrieval (CBR)** and **Knowledge Graph Retrieval (KGR)** [21]. CBR identifies semantically similar documents or passages using keyword matching, vector space models, or deep learning methods. It focuses on the intrinsic content of data, aiming to retrieve the most relevant information from large corpora. KGR, in contrast, leverages graph-structured data to model entities and their relationships. Querying these graphs provides rich, context-aware responses, especially useful when relational reasoning is required.

In the context of multimodal RAG, the retrieval module must also handle and integrate inputs from different modalities. Unlike traditional RAG systems that focus primarily on text, multimodal RAG must process and retrieve diverse content types such as images and videos. For instance, multimodal RAG models demonstrate how combining visual and textual information enhances the model's capability to understand complex real-world scenarios. The retrieval module not only identifies relevant text but also extracts meaningful features from visual content, which are then passed to the generator. This multimodal integration improves the model's understanding and enables it to generate outputs with richer semantics, as seen in tasks like image captioning, visual question answering, and multimodal dialogue systems.

The combination of advanced retrieval technologies and generative models marks a significant leap forward in AI, particularly in scenarios requiring the synthesis of complex, heterogeneous data sources. Nevertheless, challenges persist in enhancing retrieval efficiency, minimizing noise, and ensuring the seamless integration of retrieved information. Future work is expected to further optimize the interaction between retrieval and generation components, enabling the development of more powerful, accurate, and context-aware AI systems.

E. Multimodal Embedding

Multimodal embedding refers to the process of mapping data from diverse modalities—such as text [22], images, audio, and video—into a shared representation space, thereby enabling integration across heterogeneous data types. This technique lies at the core of multimodal learning, allowing systems to effectively understand, analyze, and synthesize information from multiple sources. The ability to fuse information from different modalities is increasingly important for tasks that require contextual understanding, such as multimodal retrieval, visual question answering, and multimedia content generation. By embedding diverse data types into a common space, models can reason across modalities and generate deeper, more nuanced responses to complex queries [23].

A key technique in multimodal embedding is **joint embedding** [24], which projects data from different modalities into a shared vector space. This allows the model to uncover correlations and similarities across modalities. For example, a textual description of an object and its corresponding visual representation can be mapped to proximate locations in the embedding space, making it easier for the model to align text and images. This is particularly useful for tasks like image captioning or multimodal dialogue, where the model must generate responses that are contextually grounded in both visual and textual inputs [25].

The central challenge of multimodal embedding lies in creating a unified representation that spans the diverse characteristics of various data types. While all modalities carry semantic information, they do so through fundamentally different means—text relies on syntax and semantics, images on visual features, and videos on temporal dynamics. To address this, state-of-the-art approaches employ deep neural networks: convolutional networks for visual data and Transformer-based models for text. These modality-specific features are then projected into a common embedding space through cross-modal or multi-view learning, ensuring the resulting representations are both informative and compatible across modalities.

In the context of RAG, multimodal embedding enhances the model's ability to generate context-aware outputs. By incorporating embeddings from multiple modalities—such as text and images—RAG systems can retrieve a more comprehensive set of context signals. For instance, in a multimodal RAG setting, the retriever may access not only relevant text documents but also associated visual data (e.g., images or diagrams). These visual elements are embedded alongside textual content into the same shared space, providing the generator with richer and more diverse inputs. As a result, the model can generate responses that are not only semantically precise but also grounded in visual information.

F. Multimodal Fine-Tuning

Multimodal Fine-Tuning refers to the process of optimizing a pretrained multimodal model for a specific task or domain, thereby improving its adaptability to targeted application scenarios. This approach leverages the general feature representations learned from large-scale multimodal datasets during pretraining and refines them using task-specific data. The primary goal is to enhance task-specific performance by employing **transfer learning**, which enables the transfer of knowledge from the pretrained model to the new task. This method reduces the need for extensive labeled data and accelerates convergence.

The multimodal fine-tuning process typically involves the following steps:

1) **Loading a pretrained multimodal model:** The first step involves loading a model that has been pretrained on large-scale multimodal datasets, such as image-text paired corpora [26]. These models possess strong general-purpose feature extraction capabilities and are adept at capturing cross-modal semantic alignments.

Representative examples include **CLIP (Contrastive Language-Image Pre-training)** [27], [28] and **BLIP (Bootstrapping Language-Image Pretraining)** [29], both of which have demonstrated effective semantic alignment between visual and textual modalities. The key here is to retain the general representations learned from large-scale pretraining, which are often robust and reusable across various tasks. Utilizing these models prevents the need to train from scratch, saving substantial computational resources and time.

2) **Fine-tuning on task-specific datasets:** After loading the pretrained model, the next step is to fine-tune it on data relevant to the target task. For instance, in a medical imaging and report generation scenario, each sample in the dataset may consist of a medical image and its associated textual description. The goal of fine-tuning is to adapt the model parameters—either partially or entirely—to better capture domain-specific features. During this phase, the parameters of the pretrained model are updated using the task dataset by minimizing a loss function tailored to the task (e.g., image-text matching loss). Depending on the complexity of the task, one may opt for fine-tuning only the top layers, or in some cases, the entire model. The objective is to allow the model to specialize in the task while leveraging the general representations it has previously acquired.
3) **Evaluation on validation and test sets:** Once fine-tuning is complete, the model's performance must be evaluated using a validation set and a separate test set. The validation set is typically used during training for hyperparameter tuning and model selection, while the test set assesses the model's generalization capabilities. Common evaluation metrics vary depending on the task. For medical image-text alignment, metrics such as cosine similarity, classification accuracy, and recall are often employed. If the model performs well across both validation and test sets, the fine-tuning can be considered successful.

In the medical field, multimodal RAG models can be fine-tuned to jointly interpret medical images and textual descriptions, enabling them to generate more accurate diagnostic reports or treatment suggestions, thereby assisting clinical decision-making. Beyond healthcare, multimodal fine-tuning also holds promise in other domains, such as autonomous driving (where it integrates visual and radar data) and intelligent education (where it combines text, images, and audio to deliver personalized learning experiences).

However, several challenges remain. One key issue is balancing the generalization capability of the pretrained model with the specificity of task data. Another is preventing overfitting during the fine-tuning process. Future research may explore more efficient fine-tuning strategies, such as adapter-based methods, which insert lightweight adapter modules into pretrained architectures and require updating only a small subset of parameters. This approach enables rapid adaptation to new tasks while preserving the general features of the original model.

Moreover, techniques such as **self-supervised learning** [30] can further reduce dependence on labeled data and improve model performance in low-resource scenarios. These strategies will play an important role in enhancing the flexibility and efficiency of multimodal fine-tuning across a wide range of applications.

G. Advancements and Challenges in Multimodal Embedding

- Multimodal embedding techniques have demonstrated superior perception and reasoning capabilities by integrating information across visual, linguistic, and other modalities. These models are not limited to processing single-modality data; rather, they perform complex tasks via cross-modal fusion. For instance, the BLIP-2 model achieves **bootstrapped language-image pretraining** by freezing the image encoder and the large language model, enabling efficient multimodal understanding and generation [31]. This strategy significantly reduces computational cost by limiting parameter updates during training, while still maintaining high performance on downstream multimodal tasks.
- Deep neural networks play a central role in multimodal embedding. Architectures such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are used to extract high-level features from images, text, and other data modalities. Furthermore, the rise of self-supervised learning has injected new vitality into the field. Self-supervised methods model semantic relationships across modalities without requiring labeled data, thereby reducing annotation costs and improving generalization. Notably, **contrastive learning** [32] has proven effective in enhancing the quality of multimodal representations by maximizing the similarity between positive pairs and minimizing it for negative pairs.
- **Alignment and mapping** remain critical challenges in multimodal embedding. To establish effective correspondences across modalities, researchers have proposed various alignment mechanisms. Cross-modal attention mechanisms help build attention maps between different modality features, enabling the model to capture their associations more effectively. For example, **Vision-and-Language Transformers (ViLT)** [33] jointly model visual and linguistic features, achieving efficient cross-modal alignment through a unified Transformer architecture.
- In addition, **Generative Adversarial Networks (GANs)** [34] have been widely applied to multimodal embedding tasks. In image-text alignment, GANs can generate images that correspond to text descriptions, or vice versa, thereby facilitating mutual understanding between modalities. In vision-language tasks, for instance, GANs may generate textual descriptions aligned with image content, or synthesize images based on natural language prompts. This capability of cross-modal

generation enhances the expressive power of multimodal models and enables higher-level understanding and synthesis across modalities such as text, image, and speech. GANs also contribute to data augmentation, especially under data-scarce conditions, by generating additional training samples to improve model robustness and generalizability.

- **Multi-Task Learning (MTL)** [35] plays a vital role in multimodal embedding by optimizing multiple related tasks simultaneously through shared representation learning. In systems such as **Visual Question Answering (VQA)**, which require simultaneous understanding of visual and textual inputs, MTL enables the model to jointly learn vision tasks (e.g., object detection, scene analysis) and language tasks (e.g., parsing, semantic understanding). This joint optimization fosters synergy between modalities and significantly improves overall task performance. Shared representations learned across tasks enhance the model's ability to generalize and solve complex multimodal challenges.

IV. Evaluation of Retrieval-Augmented Generation Systems

With the growing popularity of Retrieval-Augmented Generation (RAG) techniques, the evaluation methodologies for RAG systems are evolving rapidly. Traditional human evaluations typically rely on expert judgments, which are often subjective and may lead to biased or inconsistent results. Although classical information retrieval metrics have long served as standard tools to assess model performance, they face limitations in large-scale RAG applications. In particular, these metrics often struggle to accurately reflect user needs and real-world system effectiveness.

This section provides a systematic review of two mainstream RAG evaluation frameworks: **RAGAS (Retrieval-Augmented Generation Assessment System)** [36] and **ARES (Automated RAG Evaluation System)** [37].

A. RAGAS

1) Workflow: RAGAS begins by automatically generating a set of query-context-answer triplets using a large language model (e.g., GPT-3.5), grounded in a predefined knowledge base. These triplets include both positive samples (e.g., relevant context and correct answers) and negative samples (e.g., irrelevant context and incorrect answers). This method enables the construction of a diverse and balanced dataset to train evaluation models and ensure broad coverage across scenarios.

RAGAS trains independent evaluation models for each metric: **faithfulness, answer relevance**, and **context relevance**. Each model receives a query-context-answer triplet and classifies it according to its target criterion. Through contrastive learning and fine-tuning, these evaluators are trained to accurately reflect the system's real-world performance.

Once training is complete, RAGAS applies the evaluation models to assess different RAG systems. The predicted scores are compared with human annotations to validate the reliability and accuracy of the framework. Notably, RAGAS enables high-quality, scalable evaluation even without extensive manual labeling.

2) Faithfulness: Faithfulness measures whether the generated answer is grounded in the retrieved context. High faithfulness indicates that all statements in the answer can be logically inferred from the context, minimizing hallucinations. This metric is crucial in fact-sensitive tasks.

Evaluation method: RAGAS decomposes the generated answer into individual claims and uses a large language model to verify whether each claim is supported by the retrieved context. The score is computed as:

$$F = \frac{|V|}{|S|}$$

where $|V|$ is the number of verified claims and $|S|$ is the total number of claims.

3) Answer Relevance: Answer relevance assesses whether the generated answer directly addresses the query without including irrelevant or redundant content. It focuses on informativeness and specificity, regardless of factual correctness.

Evaluation method: RAGAS generates multiple related questions based on the output answer using an LLM, and computes cosine similarity between each generated question and the original query:

$$AR = \frac{1}{n} \sum_{i=1}^{n} \text{sim}(q, q_i)$$

where q is the original query, q_i is the i-th generated question, and $\text{sim}(\cdot)$ denotes cosine similarity.

4) Context Relevance: Context relevance evaluates whether the retrieved passages are concise and directly relevant to answering the query, without including unnecessary or distracting content.

Evaluation method: RAGAS extracts the most relevant sentences from the retrieved context. The fewer sentences required to answer the query accurately, the higher the context relevance. The score is given by:

$$CR = \frac{\text{Number of relevant sentences}}{\text{Total number of context sentences}}$$

5) Assessment of RAGAS: Experimental results using the WikiEval dataset demonstrate RAGAS's high accuracy and reliability. The framework achieved over 95% agreement with human annotations on faithfulness and 90% on answer relevance. While context relevance evaluation showed slight discrepancies—especially with longer texts—RAGAS tended to adopt a conservative approach, prioritizing precision.

B. ARES

ARES is designed to automatically evaluate RAG systems along three key dimensions: **context relevance**, **answer faithfulness**, and **answer relevance**.

1) Workflow: ARES leverages generative large language models (e.g., FLAN-T5 XXL) to synthesize query-answer pairs from a defined corpus. These samples include both positive (relevant documents and correct answers) and negative examples (irrelevant documents or incorrect answers), facilitating the construction of a robust training dataset.

ARES trains independent binary classifiers for each evaluation metric. Each classifier receives a query-document-answer triplet and determines whether the input meets the requirements of the corresponding metric. These models are fine-tuned using contrastive learning objectives and validated with a small set of human-annotated examples.

After training, ARES applies the evaluation models to assess various RAG systems. The predictions are used to compute overall scores, and the PPI (Posterior Predictive Interval) method is employed to calculate confidence intervals. PPI helps correct prediction errors using limited human labels, enhancing the accuracy of evaluations on large-scale unlabeled datasets.

2) Context Relevance: This metric assesses whether the retrieved paragraph is relevant to the query. A fine-tuned LLM classifies each query-paragraph-answer triplet accordingly, ensuring contextual alignment.

3) Answer Faithfulness: Answer faithfulness determines whether the generated answer accurately reflects the content of the retrieved documents and avoids unsupported reasoning. The metric is defined as:

$$\text{Answer Faithfulness Score} = \frac{\text{Number of faithful answers}}{\text{Total number of answers}}$$

A faithful answer is one that strictly adheres to the facts provided in the context.

4) Answer Relevance: This metric evaluates whether the generated answer addresses the query appropriately and aligns with the retrieved document. Similar to other metrics, a binary classifier is used to determine whether each triplet is a valid instance, ensuring full query resolution.

V. Real-World Applications

A. Applications in the Medical Domain

Medical imaging is one of the most important data sources in clinical diagnosis, including X-rays, CT scans, and MRI images. However, relying solely on image analysis can lead to misdiagnoses or missed detections, particularly in early-stage disease screening. Integrating imaging data with other medical information (e.g., electronic health records, lab results) provides a more comprehensive diagnostic view for clinicians. RAG-based multimodal embedding techniques effectively combine image and textual data, retrieving information from external medical knowledge bases to enhance model performance and reduce hallucination phenomena.

For example, in early lung cancer screening, AI models can combine CT scan data with patient history and familial risk factors to retrieve relevant medical literature and generate more accurate diagnostic suggestions. Additionally, AI systems can generate personalized treatment recommendations by leveraging multimodal information, assisting physicians in formulating more effective therapeutic plans.

Personalized medicine aims to develop customized treatment plans based on individual patient characteristics such as genomic profiles, lifestyle, and medical history. Traditional AI-based medical models often rely on standardized data and protocols, which may fail to account for individual differences. By incorporating RAG-based multimodal embedding techniques, it becomes possible to integrate genomic data, medical images, and clinical records for personalized inference. In cancer therapy, for instance, such systems can retrieve cutting-edge research based on a patient's genomic mutations, tumor imaging, and clinical profiles, thereby predicting treatment outcomes and recommending optimal plans. This enhances precision while minimizing adverse effects [38], [39].

Clinical Decision Support Systems (CDSS) [40] are vital tools that assist physicians by analyzing patient data to provide diagnostic and therapeutic suggestions. Traditional CDSS approaches are often rule-based or expert-driven, but they may be inadequate for complex cases. RAG-based multimodal systems can integrate diverse modalities (e.g., imaging, clinical notes, genomics) and access external medical knowledge to support decision-making.

For complex clinical scenarios, such systems can retrieve relevant literature, case reports, and clinical guidelines. They then align this information with patient-specific data to generate personalized recommendations.This improves diagnostic accuracy and offers comprehensive insights to support medical professionals in complex decision-making processes [41].

B. Applications in Artistic Creation

In the domain of creative arts, there is growing demand for **multimodal artistic creation** [42]. Artistic works are no longer confined to a single medium, but increasingly integrate text and images, images and audio, or even images and video. Artists can input a simple textual description to generate visually coherent and stylistically rich artwork that conveys emotional depth. With RAG, the generation process can incorporate and fuse diverse modalities, enabling cross-modal understanding and creativity.

1) Artistic Image Generation and Editing: Artistic image generation is a key application of RAG in the creative domain. Traditional generative models—such as GANs (Generative Adversarial Networks) and VAEs (Variational Autoencoders) [43]—have been used to generate stylized images from given inputs. However, these models often suffer from limitations such as lack of fine detail or style inconsistency. By integrating RAG mechanisms, external knowledge and reference styles can be injected into the generation pipeline, enhancing both the richness and coherence of outputs.

CreativeSynth exemplifies this approach by integrating multimodal inputs, particularly deep fusion of images and text, to produce high-fidelity, stylized artistic images. It employs two core mechanisms—**Aesthetic Preservation** and **Semantic**

Fusion—to mitigate common issues in artistic editing such as style distortion and semantic inconsistency [44].

2) Personalized Artistic Creation: Another critical application of RAG embedding lies in personalized art generation. Artists and designers increasingly demand tools that can generate tailored artworks with minimal input. Traditional models often struggle with high-level personalization, especially in tasks such as style transfer or identity-aware generation.

By leveraging intelligent matching and alignment between image and text inputs, artists can achieve complex creative goals with simple prompts. In style transfer tasks, for instance, CreativeSynth uses a customized **Cross-Attention Mechanism** to precisely control the interaction between textual and visual elements. This ensures that the generated artwork not only meets personal stylistic requirements but also preserves meaningful artistic depth. The system is capable of processing multiple artistic dimensions—such as tone, composition, and form—thereby producing works that reflect the artist's unique vision.

3) Enhancing Art Education: Beyond content generation, RAG technologies show strong potential in art education and appreciation. Traditional art education depends heavily on instructor-led interpretation and student perception. RAG systems can enhance this process by generating explanatory text, style analysis, and contextual background alongside artworks.

LLaVA-Docent [45] is a RAG-based tool designed for art appreciation and education. It provides users with detailed descriptions, historical context, and analytical commentary based on input queries. This enriches users' aesthetic experience. Students can better understand the cultural and creative intent behind artworks, and receive guided suggestions for their own creations. Additionally, LLaVA-Docent supports artistic composition feedback and style coaching, enabling more personalized and insightful creative instruction.

VI. Conclusion

This survey reviewed recent advancements in Retrieval-Augmented Generation techniques, particularly focusing on their integration with multimodal embedding technologies. We examined how RAG enhances text generation tasks by leveraging information from multiple modalities, significantly improving model performance in complex scenarios.

RAG systems, when equipped with multimodal embedding capabilities, demonstrate strong potential across a wide range of applications, particularly in domains such as medicine and content creation. They not only improve accuracy and efficiency but also help mitigate hallucination issues by grounding outputs in relevant, retrieved information.

Despite these advances, challenges remain—particularly in seamless cross-modal data fusion and maintaining high generation quality. Future research is expected to address these limitations, driving broader adoption of multimodal RAG across diverse real-world applications. Ultimately, this line of work is poised to open new directions for intelligent systems capable of multimodal understanding and generation.

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